

Question Responses Natural Resources Specialist

November 24, 2025

0.1 Objective

To provide program management and project analyses in support of evidence-based policy making that benefits both people and our planet.

1. Job description

The Environmental Toxins Program Specialist Senior position's purpose is to reduce toxics exposure through pathways related to the built and natural environments, and to work on Issues related to the built environment, including air quality, toxics exposure, planning related to land use, environmental justice, and emerging environmental health concerns. This position advises County leadership, partners, and residents on health-protective actions during public health emergencies related to toxic exposures.

As the Environmental Toxins Program Specialist Senior you will:

Act as the county's subject matter expert and technical lead in public health related air quality and brownfield (contaminated property) work, and advising program leadership, departmental leadership and elected officials on policy and program development of a range of politically sensitive topics

Provide assessment, consultation, community engagement, policy, project management, grants administration, technical expertise, supervision of interns, and program development and execution

Be the driver of policy and program development and execution, under the oversight of program and divisional management

Rapidly process highly technical content and skillfully deliver critical Information to a variety of internal and external stakeholders of matters of high concern to the county and its constituents

1.1 Required skills

Minimum Qualifications/Transferable Skills*:

A Bachelor's degree from an accredited college or university in Public Health, Policy and Program Development or a related field AND Five (5) years of increasingly responsible and major program development, policy analysis, or program compliance/monitoring and evaluation experience Ability to represent the Department and/or County in public and private meetings and other gatherings related to areas of the Environmental Toxins Program Effective skills in managing team/group, organizational dynamics, and consensus building.

Required: Knowledge, Skills and Abilities - Knowledge of toxics exposure through air

quality, toxics exposure, planning related to land use, environmental justice, and emerging environmental health concerns. Demonstrated knowledge of US Environmental Protection Agency air and land, and Oregon Department of Environmental Quality policies and programs. Experience as subject matter expert and technical lead in public health actions and program development related to air quality and brownfield (contaminated property) work. Experience with analysis and development of proposed and current federal, state, regional and local policy, programs and legislation pertaining to air and land quality.

Preferred Qualifications/Transferable Skills*: You do not need to have the following preferred qualifications/transferable skills to qualify. However, keep in mind we may consider some or all of the following when identifying the most qualified candidates. Please clearly explain on your application how you meet any of the following preferred qualifications/transferable skills.

Knowledge and experience with environmental justice, health equity and the built environment.

I. Accept the info

II. How has your education and/or experience prepared you to provide technical consultation work related to health and the built environment to stakeholders, including County leadership and residents?

I hold an M.S. degree in environmental chemistry and was awarded a fixed price contract with the U.S. Army Corps of Engineers for a study of trace metal and organochlorine pesticide contamination of Cochiti Reservoir in northern New Mexico. I spent nearly a decade working as an environmental regulatory compliance specialist primarily under RCRA and NEPA. All of my projects at that time were concerned with human and environmental health. Many projects were directly related to toxics exposure through pathways related to the built and natural environments. For example, I was project lead for a team of health physicists and environmental toxicologists given the task of creating a methodology to compare chemical and radionuclide risks to human health ("A Method for Comparative Assessment of Chemical and Radionuclide Risks"). The methodology was used in support of human health evaluations at the Waste Isolation Pilot Plant. I wrote the section on radiological monitoring of foodstuffs for the Environmental Monitoring Plan in support of Los Alamos National Lab's compliance with DOE Orders 5400.1 and 5400.5. And I wrote comment responses for the Air Quality section of the White Sands Missile Range Draft Environmental Impact Statement.

More recently I was contracted by the Oregon Environmental Council to research and write a white paper on reactive nitrogen in Oregon. The paper made recommendations for mitigating the adverse effects of reactive nitrogen (Nr) to human and environmental health in Oregon, both now and in the future. The immediate, near-term goal of the recommendations was to reduce the risks to human and environmental health that are associated with Nr. The long-term goal was to ensure

that there are abundant and high-quality freshwater resources available for future generations of Oregonians.

Finally, I have spent the last six years updating and augmenting my skill set to better support my career in public service. I earned a second M.S. in systems science plus graduate certificates in sustainability and in statistics. I was awarded an NSF IGERT fellowship in the Earth, Environment, and Society doctoral program at Portland State University for research in ecosystem services. Specifically, I was developing the theoretical underpinnings of an agroecological paradigm for sustainable agriculture. Ecologically resilient foodsheds that produce high quality foods without excessive petrochemical inputs are, of course, desirable for the long-term health of both people and planet.

I am aware of the detailed and rigorous EPA requirements and protocols for conducting a human health risk assessment (planning, hazard identification, dose-response assessment, exposure assessment, risk characterization). I have had some training in risk assessment. Environmental toxicology, especially the fate and transport of chemicals, was part of my MS training in environmental chemistry.

- III. Describe a project or initiative where you applied an equity or environmental justice lens or framework. How did applying this lens or framework impact the end results or your project or initiative?

The environmental justice movement brings attention to the full range of conditions that surround people in their daily lives. While initially focused on the existence of inequity in the distribution of environmental risks, the environmental justice movement has expanded as an organizing frame to include civil rights, immigrant rights, indigenous rights, the labor movement, climate justice, as well as growing concerns around the systems of food and energy production. Ultimately, as Doreceta Taylor points out, “the environmental justice movement is transformative, seeking broad and sweeping changes in the social structure and its ideological foundations.” (Dorceta E. Taylor. “The rise of the environmental justice paradigm: Injustice framing and the social construction of environmental discourses”. In: *The American Behavioral Scientist*, 43.4 (Jan. 2000), pp. 508–580.)

I believe that environmental justice is clearly a key requirement for environmental, economic, and social sustainability. And by logical implication, the tripartite definition of sustainability produces human health. While I have not explicitly applied an environmental justice lens to my projects, I fully recognize the importance of such a lens. Environmental justice was a key component in the course work for the graduate certificate in sustainability as well as for the course work in ecosystem services through the IGERT program at Portland State University. I might also mention my absolute support of the UN Millennium Development Goals, which I believe support not only the development of substantive freedoms as defined by Nobel laureate Amartya Sen, but are the backbone of human health.

IV. Briefly describe your relevant experience related to understanding and communicating the health impacts of the built environment

I spent nearly a decade as an environmental regulatory compliance specialist steeped in federal and state regulations. I know well how to research and apply regulatory intent to real-world situations when dealing with issues regarding hazardous waste, air quality, and water quality and discharge permits (under 40 CFR Subchapters C, D, I) as well as with chemicals (under 40 CFR Subchapters E and J). I regularly wrote responses to questions on environmental compliance from clients and the general public regarding hazardous and solid waste disposal, action limits for chemicals under RCRA and CERCLA and legal requirements under the Clean Water Act and the Clean Air Act. I evaluated recognized environmental conditions in conjunction with Phase I environmental site assessments and communicated those findings to the clients and the general public. I would also mention that I spent nearly four years teaching high school chemistry and can readily explain highly complex concepts to audiences with limited technical background.

Developed recommended method detection limits, practical quantitation limits, and spiking amounts and designed an evaluation report format for the Mixed Waste Methods Laboratory (MWML) at Los Alamos National Laboratory (LANL)(the MWML was tasked with assessing the feasibility of using EPA SW-846 methods for analysis of TRU mixed waste); designed a series of experiments for the Idaho National Engineering Laboratory (INEL) to determine if theoretical methods can be used to predict flammability of headspace volatile organic compounds; wrote sections on facility description, boundary conditions and process tolerance limits, waste shipment screening, quality assurance and quality control for the revised Waste Isolation Pilot Plant (WIPP) Waste Analysis Plan (WAP); developed sampling and analysis plans for hazardous and low-level mixed wastes for personnel of LANL's Chemical and Mixed Waste Operations Section (CMWOS)(the documents serve CMWOS personnel as practical supplements to the waste analysis plan in LANL's RCRA facility operating permit); compiled technical contributions by staff members to generate a method for the comparative assessment of chemical and radionuclide risks for the WIPP; evaluated the hazardous constituent source term potential from RCRA solid waste management units (SWMUs) undergoing environmental restoration at LANL's TA-54 (the evaluation was made by applying statistical models to an extensive dBase IV database compiled from LANL hazardous waste disposal records); prepared the technical approach, including all physical chemistry calculations, for estimating the gas-phase, source term amounts of volatile organic compounds (VOCs) found in TRU mixed waste that is to be sent to the WIPP (the gas phase VOC source term was used by Sandia National Laboratories (SNL) in modeling for the long-term performance assessment of the WIPP repository); wrote an analytical procedure (flame test) for evaluating the flammability of headspace samples from TRU waste containers that are to be sent to the WIPP (the procedure was developed to comply with EPA's, "Notice of Conditional No-Migration Determination for the WIPP (NMD) (FR 47700)," and is part of the "WIPP Waste Characterization Program Sampling and Analysis Guidance Manual (Guidance

Manual)," DOE/WIPP 91-043); wrote the portion of the position paper that defined DOE's RCRA compliance approach to the flammability requirements promulgated in the EPA's NMD; wrote the section on radiological monitoring of foodstuffs for the Environmental Monitoring Plan (EMP) submitted to the DOE by the LANL Environmental Protection Group (EM-8)(the EMP is designed to demonstrate LANL compliance with DOE Orders 5400.1 and 5400.5); wrote an operating instruction for chemical, gas cylinder, and mixed waste segregation and storage at LANL's TA-54, Area L (this document ensures that the RCRA hazardous and mixed wastes stored at TA-54 are managed in compliance with LANL's RCRA facility operating permit); wrote a standard operating procedure for the construction of pits and shafts for the personnel of LANL's Radioactive Waste Operations Section; wrote the executive summary of the "Quality Assurance Program Plan for the Waste Isolation Pilot Plant Experimental-Waste Characterization Program," DOE/EM/48063-1; and wrote a standard operating procedure for the activities of the decontamination and decommissioning group at LANL's TA-50. Technical Presenter - Presented a paper entitled, "A Method for Comparative Assessment of Chemical and Radionuclide Risks" at the American Nuclear Society Summer Meeting in San Diego, CA.